Information Retrieval

Retrieval Models III: DfR and Pagerank

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Retrieval Models III: Language model and Pagerank

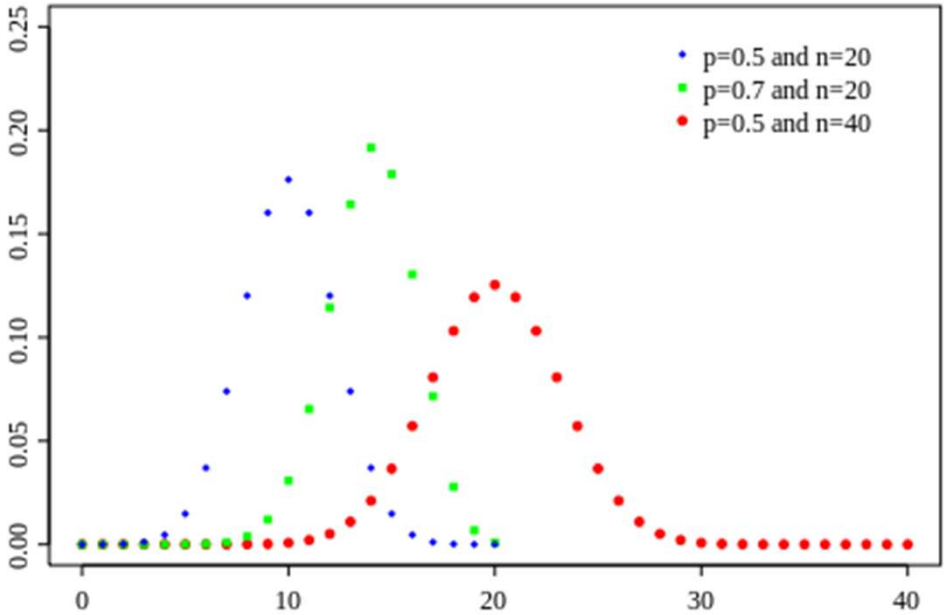
Roadmap the lecture:

- Evaluation continued
- Divergence from Randomness model
- Pagerank

Binomial Distribution

Binomial distribution is the discrete probability distribution of the number of successes in a sequence of *N* independent *yes/no* experiments

- Each of which yields success with probability *p*.
- Such a success/failure experiment is also called a Bernoulli experiment or Bernoulli trial;
- When *N* = 1, the binomial distribution is a Bernoulli distribution.
- The binomial distribution is the basis for the popular binomial test of statistical significance.



Binomial Distribution

An example

$$P(n) = \binom{N}{n} \cdot p^n \cdot (1-p)^{N-n}$$

Imagine you go on a sailing trip on the East Coast. Every second day, there is a beautiful sunset, i.e. $p = 1/2$. You go sailing for a week ($N = 7$). What is your chance to have exactly three ($n = 3$) beautiful sunset?

Binomial Distribution

Just another example

$$P(n) = \binom{N}{n} \cdot p^n \cdot (1-p)^{N-n}$$

In your *company, 2 percent of the products delivered have a failure, i.e. p* = 2/100. Per day, you deliver 100 products (boats) (*N* = 100). In average, you expect 2 boats per day to be faulty. What is the probability that exactly one boat is faulty?

Divergence from Randomness (DFR)

- Basic idea: ”The more the divergence of the within-document term frequency from its frequency within the collection, the more divergent from randomness the term is, meaning the more the information carried by the term in the document.”

$$weight(t \mid d) \propto -\log P_M(t \in d \mid collection)$$

M stands for the type of model of the divergence from randomness employed to compute the probability.

In the next slide, the binomial distribution (*B*) is used as the model of the divergence from randomness.

See http://ir.dcs.gla.ac.uk/terrier/doc/dfr_description.html

Divergence from Randomness (DFR)

Binomial Distribution as Randomness Model

- *TF* - Term frequency of term *t* (occurrence of *t*) in the collection
- *tf* - Term frequency of term *t* in the document *d*
- *p* - Probability to draw a document (*p* = 1/*N*, *N* is number of documents)

$$-\log P_B(t \in d \mid collection) = -\log \binom{TF}{tf} \cdot p^{tf} \cdot (1-p)^{TF-tf}$$

The probability that

- the event (that occurs with probability *p*) occurs *tf* times in *TF* trials
- a document occurs *tf* times in *TF* trials
- a sunny day (which occurs with 1/*N*) occurs on *tf* days in a *TF* days holiday

Divergence from Randomness (DFR)

Binomial Distribution as Randomness Model

$$-\log P_B(t \in d \mid collection) = -\log \binom{TF}{tf} \cdot p^{tf} \cdot (1-p)^{TF-tf}$$

- If N is the number of documents, then $TF/N = \lambda$ is the average occurrence of term t.
- The above is minimum for $tf = \lambda$, meaning that term t has a random distribution.
- its distribution does not diverge from randomness
- and as such is not informative.

Divergence from Randomness (DFR)

Binomial Distribution as Randomness Model

Summary:

- Term-weights: measuring the divergence between a term distribution produced by a random process and the actual term distribution.
- Term-weight are inversely related to the probability of term-frequency within the document d obtained by a model of randomness
- There are many ways to choose the model of randomness, each of these provides a basic DFR model

Page rank

Introduction

- The heart of Google's searching software is PageRank™, a system for ranking web pages developed by Larry Page and Sergey Brin at Stanford University
- Essentially, Google interprets a link from page v to page u as a vote, by page v, for page u.
- BUT these votes doesn't weigh the same, because Google also analyzes the page that casts the vote.

- The authority of a page is the sum of the authorities of the pages that reference the page.

Page rank

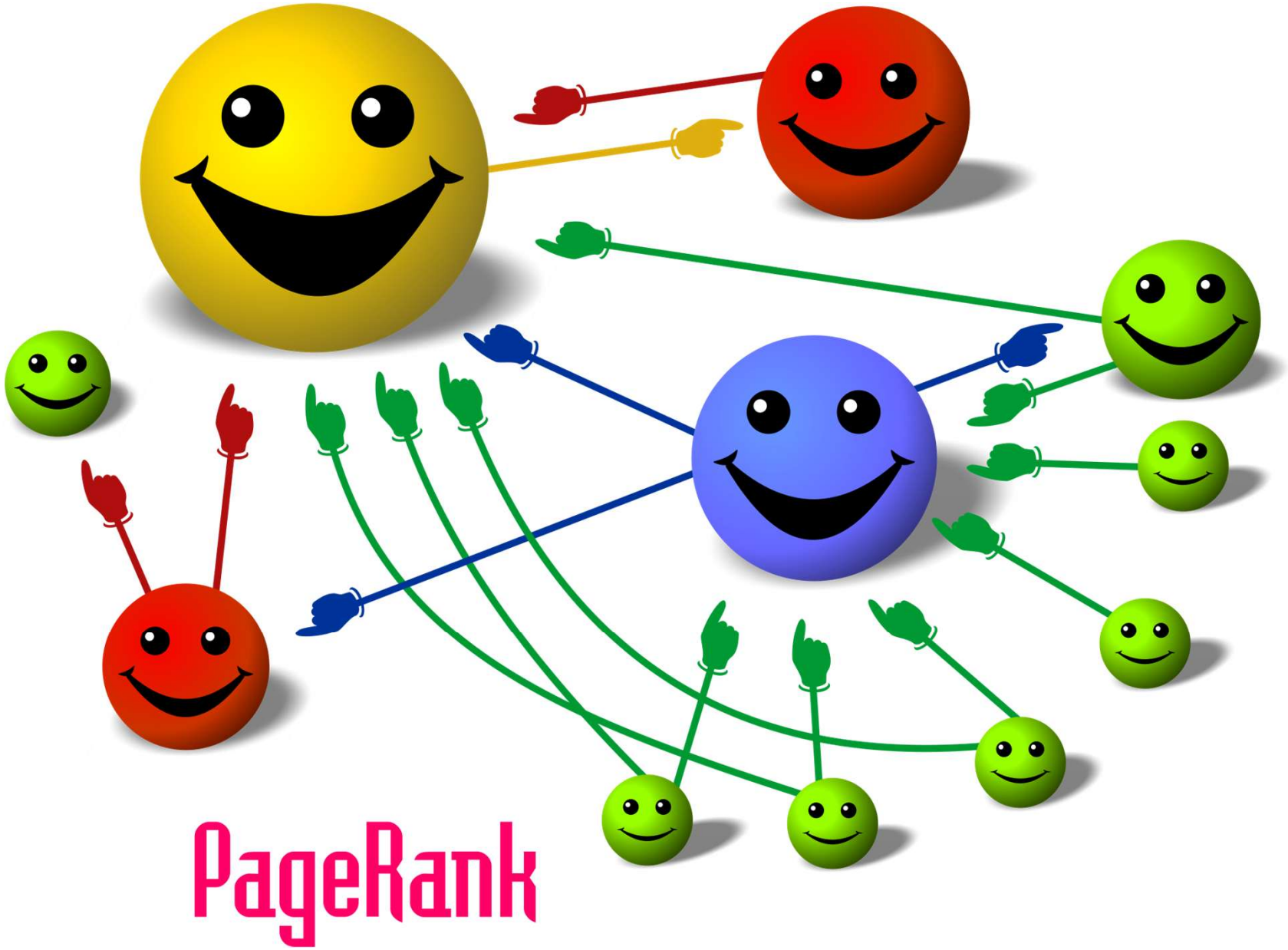
Introduction

PR	page rank reflecting its "authority", which can be used to rank pages
u, v	two pages
U	set of pages
N(*v*)	number of outgoing links in page *v*
d	a damping factor which can be set between 0 and 1

$$PR(u) := (1-d) + d \cdot \sum_{v \to U} \frac{PR(v)}{N(v)}$$

- the summation is over the set of pages *v* that have a link to page *u*
- the page rank of page *u* is recursively defined by the page rank of those pages which link to page *u*

The size of each face is proportional to the total size of the other faces which are pointing to it.



Page rank

The Random Surfer Model

- Page rank is considered as a model of user behaviour, where a surfer clicks on links at random with no regard towards content.
- The random surfer visits a web page with a certain probability which derives from the page's page rank
- The probability that the random surfer clicks on one link is solely given by the number of links on that page.
- One page's page rank is divided by the number of links on the page.

Page rank

The Random Surfer Model

- So, the probability for the random surfer reaching one page is the sum of probabilities for the random surfer following links to this page.
- This probability is reduced by the damping factor *d*.
- The justification within the Random Surfer Model, therefore, is that
  - the surfer does not click on an infinite number of links,
  - but gets bored sometimes and jumps to another page at random.

Page rank

The damping factor *d*

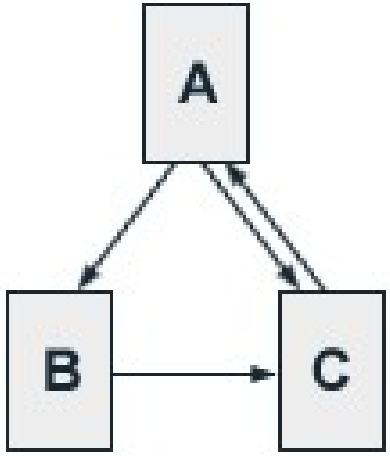
- The probability for the random surfer not stopping to click on links is given by the damping factor *d*,
  - depends on probability
  - set between 0 and 1
- The higher *d* is, the more likely will the random surfer keep clicking links.
- Since the surfer jumps to another page at random after he stops clicking links, the probability therefore is implemented as a constant (1-*d*) into the algorithm.

Page rank

The damping factor *d*

- Regardless of inbound links, the probability for the random surfer jumping to a page is always (1-*d*)
- so a page has always a minimum page rank
- The extend of page rank benefit for a page by another page linking to it is reduced.

Page rank



An example

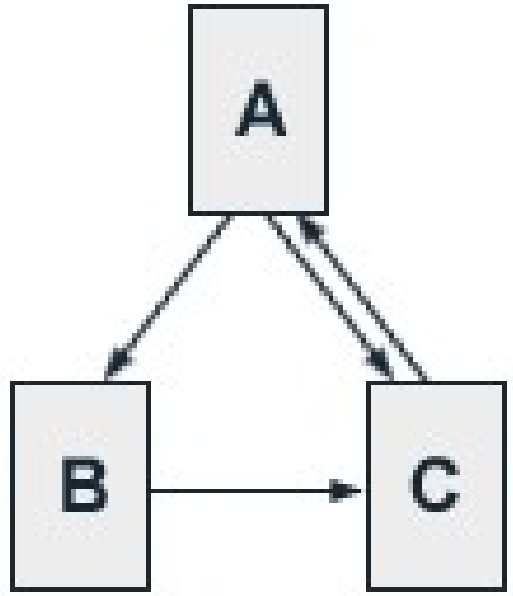
- A small web consisting of three pages A, B and C
- Page A links to the pages B and C, page B links to page C and page C links to page A
- The damping factor *d* is usually set to 0.85, but to keep the calculation simple we set it to 0.5.

The sum of all pages' PR is 3, equals the total number of web pages.

Page rank

An example

- A small web consisting of three pages A, B and C
- Page A links to the pages B and C, page B links to page C and page C links to page A
- The damping factor *d* is usually set to 0.85, but to keep the calculation simple we set it to 0.5.
  - PR(A) = 0.5 + 0.5 PR(C)
  - PR(B) = 0.5 + 0.5 (PR(A) / 2)
  - PR(C) = 0.5 + 0.5 (PR(A) / 2 + PR(B))



We get the following page rank values for the single pages.

- PR(A) = 1.07692308
- PR(B) = 0.76923077
- PR(C) = 1.15384615

The sum of all pages' PR is 3, equals the total number of web pages.

Page rank

The iterative computation of page rank

- For the simple three-page example it is easy to solve the according equation system to determine page rank values.
- In practice, the web consists of billions of documents and it is not possible to find a solution by inspection.
- Google search engine uses an approximating, iterative computation of page rank values.
  - Each page is assigned an initial starting value
  - The page rank of all pages are calculated in several computation circles based on the equations determined by the page rank algorithm.
- The iterative calculation shall again be illustrated by the three-page example, whereby each page is assigned a starting page rank value of 1.

Page rank

The iterative computation of page rank - example

Iteration	PR(A)	PR(B)	PR(C)
0	1	1	1
1	1	0.75	1.125
2	1.0625	0.765625	1.1484375
3	1.07421875	0.76855469	1.15283203
4	1.07641602	0.76910400	1.15365601
5	1.07682800	0.76920700	1.15381050
6	1.07690525	0.76922631	1.15383947
7	1.07691973	0.76922993	1.15384490
8	1.07692245	0.76923061	1.15384592
9	1.07692296	0.76923074	1.15384611
10	1.07692305	0.76923076	1.15384615
11	1.07692307	0.76923077	1.15384615
12	1.07692308	0.76923077	1.15384615

Page rank

Implementation - Dangling Links

- Links that point to any page with no outgoing links
- Most are pages that have not been downloaded yet
- Affect the model since it is not clear where their weight should be distributed
- Do not affect the ranking of any other page directly
- Can be simply removed before page rank calculation and added back afterwards

Page rank

Implementation

- Convert each URL into a unique integer and store each hyperlink in a database using the integer IDs to identify pages
- Sort the link structure by ID
- Remove all the dangling links from the database
- Make an initial assignment of ranks and start iteration
  - Choosing a good initial assignment can speed up the page rank
- Adding the dangling links back.

Page rank

Searching - Two search engines:

- Title-based search engine
  - Searches only the “Titles”
  - Finds all the web pages whose titles contain all the query words
  - Sorts the results by page rank
  - Very simple and cheap to implement
  - Title match ensures high precision, and page rank ensures high quality
- Full text search engine
  - Examines all the words in every stored document and also performs page rank (Rank Merging)
  - More precise but more complicated

Page rank

Summary

- Page rank is a global ranking of all web pages based on their locations in the web graph structure
- Page rank uses information which is external to the web pages – backlinks
- Backlinks from important pages are more significant than backlinks from average pages
- The structure of the web graph is very useful for information retrieval tasks.